Respuestas

1.-

$$T = 150 + 273 = 423\ K$$
$$P = 10\ atm$$
$$V = \ ¿$$
$$V = \frac{20 \times 0{,}082 \times 623}{3 \times 20} = 17{,}028$$
$$V_{He} = \frac{17{,}028}{4} = 5{,}676\ l$$
$$m = \frac{P \times V \times M}{R \times T} = \frac{10\ atm\ \times 4\frac{g}{mol} \times 5{,}676\ L}{0{,}082\frac{atm - L}{mol - k} \times 423K} = 6{,}5g$$

2.-

$$V = 15L$$
$$T = 25 + 273 = 298\ K$$
$$P = 700 - 23{,}8 = 676{,}2\ \ torr$$
$$n = \frac{P \times V}{R \times T} = \frac{676{,}2\ torr\ \times 15\ L}{62{,}4\ \frac{torr - L}{mol - K} \times 298\ K} = 0{,}5454\ mol\ CO_2$$

$$0{,}5454\ mol\ CO_2 \times \frac{1\ mol\ CaCO_3}{1\ mol\ CO_2} \times \frac{100\ g\ CaCO_3}{1\ \ mol\ CaCO_3} \times \frac{100\ g\ caliza}{85\ g\ CaCO_3} = 64{,}16\ g\ caliza$$

3.-

$$2L\ sln1 \times \frac{1{,}5\ eq - mol\ H_2SO_4}{1\ L\ sln1} \times \frac{1\ mol\ H_2SO_4}{2\ eq - mol\ H_2SO_4} \times \frac{98g\ H_2SO_4}{1\ mol\ \ H_2SO_4} \times \frac{100\ g\ H_2SO_4}{77{,}77\ g\ H_2SO_4}$$
$$\times \frac{1\ ml\ sln2}{1{,}027\ g\ H_2SO_4} = 184{,}05\ ml\ sln2$$

4.-

$x = g\ KCl$
$y = g\ KBr$ $\Big\}$ $x + y = 2.000$ (α)

Las reacciones químicas

$2KCl + H_2SO_4 = K_2SO_4 + 2HCl$

$2KBr + H_2SO_4 = K_2SO_4 + 2HBr$

(x g de KCl produce m_1 de K_2SO_4; y g de KBr produce m_2 de K_2SO_4)

$M(KCl) = 39.09 + 35.45 = 74.54\ g/mol$

$M(KBr) = 39.09 + 79.90 = 118.9\ g/mol$

$M(K_2SO_4) = 2 * 39.09 + 32.06 + 4 * 15.99 = 174.2\ g/mol$

$m_1 + m_2 = 1.982$

$$m_1 = x\ g\ KCl \times \frac{1\ mol\ KCl}{74.54\ g\ KCl} \times \frac{1\ mol\ K_2SO_4}{2\ mol\ KCl} \times \frac{174.2\ g\ K_2SO_4}{1\ mol\ K_2SO_4}$$

$m_1 = 1.168x$

$$m_2 = y\ g\ KBr \times \frac{1\ mol\ KBr}{118.9\ g\ KBr} \times \frac{1\ mol\ K_2SO_4}{2\ mol\ KBr} \times \frac{174.2\ g\ K_2SO_4}{1\ mol\ K_2SO_4}$$

$m_2 = 0.7325y$

Resolviendo con calculadora el sistema:

$$\left.\begin{array}{l} x + y = 2.000 \\ 1.168x + 0.7325y = 1.982 \end{array}\right\} x = 1.1871$$

$$KCl\% = \frac{x}{2} \times 100 = \frac{1.1871}{2} \times 100 = 59.35 = 59.4\%\ \therefore D$$

5.-

$$n_1 = 43.0\ g\ O_2 \times \frac{1\ mol\ O_2}{32\ g\ O_2} = 1.34\ mol\ O_2$$

$$n_2 = 5.00\ g\ He \times \frac{1\ mol\ He}{4\ g\ He} = 1.25\ mol\ He$$

$$n_3 = 11.0\ g\ N_2 \times \frac{1\ mol\ N_2}{28\ g\ N_2} = 0.39\ mol\ N_2$$

$n_T = n_1 + n_2 + n_3$

$n_T = 1.34 + 1.25 + 0.39 = 2.98\ mol$

$P_T = \dfrac{n_T R T}{V}$ $\quad T = 86°F = 30°C = 303\ K$

$$P_T = \frac{2.98\ mol \times 0.082\ \frac{atm\,L}{mol\,K} \times 303\ K}{5L}$$

$P_T = 14.8\ (atm)\ \therefore \boxed{C}$